Real Determinacy Tests for DSGE Estimation

Matyas Farkas
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Abstract

DSGE estimation usually rejects parameter draws for which the linearized model has no unique stable solution. This paper shows that one common rejection rule is too strict: a draw should not be rejected only because a complex-QZ calculation returns a complex-valued policy matrix. In a real linear rational-expectations model, stable complex roots come in conjugate pairs and describe damped cycles in real variables. If the stable-root count and the state-to-jump mapping condition hold, the policy rule is real. In one million loose-prior draws from the benchmark HLT model, the complex-QZ rule rejects 464,685 draws that real QZ accepts with median residual 8.8×10^{-14}. A Cayley real-part rule matches real QZ in this audit; doubling accepts 84,606 draws that real QZ classifies as non-unique.

Keywords: DSGE estimation; determinacy; generalized Schur; Bayesian posterior support; GPU computing.

1 Introduction

A dynamic stochastic general equilibrium (DSGE) model describes how economic variables move over time in response to shocks. In Bayesian DSGE estimation, each parameter draw must pass a solution test before its likelihood is used. The standard first-order test is determinacy: the linearized model must have one stable equilibrium path for each initial state. If the test is wrong, the posterior is wrong because valid parameter draws are either excluded or invalid draws are included.

Formally, the posterior density is proportional to

$$p(\theta \mid Y) \propto L(Y \mid \theta)\pi(\theta)\mathbf{1}\{\theta \in \Theta_D\},$$

where θ is the parameter vector, Y is the data, $L(Y \mid \theta)$ is the likelihood, $\pi(\theta)$ is the prior, and Θ_D is the set of determinate parameter values. The indicator is not a harmless numerical detail. It defines the parameter region over which the posterior is computed.

This paper proves a narrow point about that indicator. A complex-valued intermediate result from a complex-QZ calculation is not enough to reject a draw. QZ, also called the generalized Schur decomposition, is the standard algorithm for separating stable and unstable generalized eigenvalues in linear rational-expectations models (Blanchard and Kahn, 1980; Klein, 2000; Sims, 2002). Complex arithmetic can appear inside that calculation even when the economic solution is real. The valid rejection criteria are the stable-root count, the ability to write jump variables as a function of states, and the residual of the recovered policy rule.

The distinction matters for GPU code. Full generalized QZ is difficult to run efficiently in JAX on GPUs. Faster alternatives, such as Cayley transforms or doubling iterations, are useful only if they make the same accept/reject decision as real QZ on the parameter region used in estimation. The audit below shows that the old complex-QZ rule falsely rejects many valid draws in the HLT benchmark model. It also shows that doubling is too loose as a final determinacy test.

2 Objects and Definitions

After linearization, a DSGE model can be written as a generalized eigenvalue problem. Let $(D(\theta), E(\theta))$ be the real matrix pencil at parameter value θ. Its generalized eigenvalues solve

$$\det(D(\theta) - \lambda E(\theta)) = 0.$$

Finite roots with $|\lambda| < 1$ are stable: repeated application of the linearized dynamics makes them decay. Infinite generalized roots can also appear when the denominator in the generalized eigenvalue ratio is zero. They are not counted as finite stable roots.

The variables are ordered into states and jumps. State variables are fixed when current choices are made, for example capital or lagged shocks. Jump variables can adjust immediately, for example prices or interest rates. Let n_s be the number of state variables. A first-order solution is a matrix G such that

$$y_t = Gx_t,$$

where x_t is the vector of states and y_t is the vector of jump variables.

The stable generalized eigenvectors span the stable deflating subspace. Write one basis for that subspace as

$$Z = \begin{bmatrix} Z_x \\ Z_y \end{bmatrix},$$

where Z_x corresponds to states and Z_y corresponds to jumps. The graph condition says that Z_x is nonsingular. In plain language, the stable subspace must give exactly one jump-vector choice for each state vector. When the graph condition holds, the policy matrix is $G = Z_y Z_x^{-1}$.

The determinacy set used in estimation is therefore

$$\Theta_D = \{\theta : \#\{\text{finite } \lambda_i(\theta) : |\lambda_i(\theta)| < 1\} = n_s,\ Z_x(\theta) \text{ is nonsingular},\ r(\theta) \le \tau\}.$$

Here $r(\theta)$ is the relative residual of the recovered policy rule, and τ is the numerical tolerance. A rule for accepting or rejecting draws is valid on a parameter region only if it gives the same indicator as $\mathbf{1}\{\theta \in \Theta_D\}$ on that region.

Assumption 1 (real regular pencil)**.** *For the parameter vector under consideration, D and E are real, the pencil (D, E) is regular, and no finite generalized eigenvalue lies on the unit circle.*

Assumption 2 (state-to-jump graph)**.** *The stable deflating subspace has dimension n_s. For any full-rank basis $Z = (Z_x', Z_y')'$ of this subspace, the state block Z_x is nonsingular.*

3 Claim and Proof

Proposition 1 (complex stable roots imply real damped cycles)**.** *Under Assumptions 1 and 2, the first-order policy rule*

$$y_t = G x_t, \qquad G = Z_y Z_x^{-1},$$

is real. Complex finite generalized eigenvalues with modulus below one do not imply instability. If a complex-QZ implementation returns a policy matrix with non-negligible imaginary entries despite the correct stable-root count and state-to-jump graph, the complex part is a numerical extraction problem, not by itself evidence that $\theta \notin \Theta_D$.

The key idea is that a real model maps real states into real choices. A complex root is a convenient way to describe a cycle. Its conjugate root appears with it. Together the two roots describe a real two-dimensional rotation that shrinks over time when their modulus is below one.

Proof. Because D and E are real, conjugation preserves generalized eigenpairs. If $Dv = \lambda Ev$, then $D\bar{v} = \bar{\lambda} E\bar{v}$. Thus non-real generalized eigenvalues and their eigenvectors occur in conjugate pairs. The stable set $\{\lambda : |\lambda| < 1\}$ is closed under conjugation, so the stable deflating subspace is also closed under conjugation.

Any conjugation-closed complex subspace has a real basis. For each non-real vector $v = a + ib$, replace the pair $(v, \bar{v})$ with the real vectors $a = \operatorname{Re} v$ and $b = \operatorname{Im} v$. These two real vectors span the same plane as the complex pair. Repeating this replacement gives a real full-rank basis $Z = (Z_x', Z_y')'$ for the stable subspace.

Assumption 2 says that Z_x is nonsingular. The stable subspace is therefore the graph of

$$G = Z_y Z_x^{-1}.$$

Both matrices in this expression are real, so G is real. The result does not depend on the particular basis. If $\tilde{Z} = ZT$ is another full-rank complex basis for the same subspace, then

$$\tilde{Z}_y \tilde{Z}_x^{-1} = Z_y T (Z_x T)^{-1} = Z_y Z_x^{-1}.$$

In exact arithmetic, every correct basis gives the same real policy matrix.

Finally, a complex stable root $\lambda = a + ib$ corresponds to a real 2×2 block with eigenvalues $a \pm ib$. Powers of that block decay at rate $|\lambda|$. The path can oscillate, but it converges. This is stability in the real state space. □

4 Audit on a Benchmark DSGE Model

The proof says that complex stable roots are not a valid rejection reason. The next question is numerical: how often does the old rejection rule matter? I answer this with one million loose-prior parameter draws from the HLT benchmark model. HLT is used here as the model label in the replication code; the proof above does not rely on special features of this model.

Each draw is classified four ways. Real QZ is the reference calculation. Complex QZ is the old rule that rejects complex-valued extracted policy matrices. The Cayley real-part rule is a faster screen designed for batched linear algebra. Doubling is a fast iterative solver. The audit file is `benchmarks/results/qz_real_cayley_hlt_loose_1m_stream.json`.

Table 1: Accept/reject decisions in one million loose-prior HLT draws

Classification event	Count	Share
Real QZ accepts: unique stable solution	702,789	70.3%
Real QZ rejects: indeterminate	256,850	25.7%
Real QZ rejects: no stable solution	40,361	4.0%
Complex-QZ accepts: unique stable solution	238,104	23.8%
Complex-QZ rejects as "complex solution"	464,685	46.5%
Real QZ accepts those "complex" cases	464,685	46.5%
Cayley real-part disagrees with real QZ	0	0.0%
Doubling accepts a real-QZ non-unique case	84,606	8.5%

Notes: The median real-QZ residual among accepted draws is 8.8×10^{-14}; the maximum is 1.0×10^{-11}. The Cayley real-part rule matches real QZ in this audit. This is evidence on the tested HLT support, not a theorem for every DSGE pencil.

Table 1 shows that the old complex-QZ rule changes the accepted parameter region. It rejects 464,685 draws because the extracted policy matrix is complex. Real QZ accepts all of these draws and rejects none of the draws that complex QZ labels unique. The old rule therefore removes many real, stable, unique solutions.

The table also shows why fast solvers still need verification. Doubling finds all real-QZ unique cases, but it also accepts 84,606 cases that real QZ classifies as non-unique. Doubling can be useful as a fast intermediate step, but it should not be the final determinacy test. In this audit, the Cayley real-part rule agrees with real QZ on the accept/reject decision.

5 Implications for Estimation

The practical rule is simple. First, count the finite stable roots. Second, check that the stable subspace defines one jump vector for each state vector. Third, recover a real policy matrix and verify the residual. A complex intermediate matrix should trigger a real reconstruction or a fallback to real QZ. It should not by itself reject the parameter draw.

This rule is useful for GPU estimation. JAX can run Cayley-style screens and batched residual checks on the GPU. Real QZ should remain the reference validator until a reliable GPU implementation of generalized QZ is available. Doubling can still speed up the pipeline, but only behind a verification step.

6 Conclusion

Complex roots in a real DSGE model describe damped cycles, not complex-valued economic choices. The valid determinacy test is the stable-root count, the state-to-jump mapping condition, and the residual of the recovered policy rule. In the HLT audit, replacing the old complex-QZ rejection rule with real QZ changes the number of accepted draws from 238,104 to 702,789. This change is large enough to alter posterior estimates.

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References

[1] Olivier Jean Blanchard and Charles M. Kahn. 1980. The solution of linear difference models under rational expectations. *Econometrica* 48(5): 1305–1311.

[2] Paul Klein. 2000. Using the generalized Schur form to solve a multivariate linear rational expectations model. *Journal of Economic Dynamics and Control* 24(10): 1405–1423.

[3] Christopher A. Sims. 2002. Solving linear rational expectations models. *Computational Economics* 20: 1–20.

[4] LAPACK Working Group. 1999. *LAPACK Users' Guide*. Third edition. SIAM.